

Real Estate WhatsApp AI CRM Automation

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1. Executive Summary & Project Vision

This project aims to establish a robust, fully automated, and AI-assisted lead management ecosystem. By seamlessly integrating n8n orchestration, localized or cloud-based Large Language Models (LLMs), and pCloud storage, the system captures leads across fragmented platforms, automates multi-tiered nurturing based on real-time client intent, and manages secure document distribution from initial touchpoint to final closing.

2. Phased Workflow Architecture

Phase 1: Lead Inflow & Filtering

- **Multi-Channel Aggregation:** Leads originating from Email, WhatsApp, Property websites, the Bentongland website, Instagram, Facebook, YouTube, Xiaohongshu, and WeChat will be unified into the classification workflow.
- **Two-Number Routing Logic:** An AI Customer Service sub-number (011-63044931) serves as the frontline handler for greetings, preliminary FAQs, and instant property info distribution via pCloud. Once a lead is qualified, data is bridged to the Main Number (011-65144951) for high-touch conversion and volume management.
- **Initial Classification:** The AI will classify leads into distinct categories: Buyer, Seller, Buyer + Seller, Non-Agent (Broker), Agent (segmented by area), Tenant, or Other.

Phase 2: Before Viewing (AI Pre-Nurturing)

- **Trigger & Sync:** Sales personnel manually initiate this phase via a system button, which automatically synchronizes the appointment to Google Calendar.
- **Intent Tagging:** The AI sub-number applies WhatsApp Labels to categorize buyers:
 - **Hot:** High intent, scheduled for follow-up every 1-7 days.
 - **Warm:** Interested but observing, scheduled for follow-up every 7-14 days.
 - **Cooling:** Low intent, scheduled for follow-up every 2 weeks with greeting messages or similar property recommendations to prevent churn.

Phase 3: After Viewing (Automated Follow-Up)

- **Document Dispatch:** Immediately following a viewing, the n8n workflow triggers and sends the Customer Property Acknowledgement Letter to officially confirm the viewing record and communication.
- **Accelerated Follow-Up Matrix:** Post-viewing follow-ups operate on a tightened schedule to secure feedback and prompt the next action:
 - **Hot:** Follow-up must be completed within 24 - 72 hours.
 - **Warm:** Follow-up must be completed within 3 - 7 days.
 - **Cooling:** Scheduled for periodic maintenance after 7 days.

Phase 4: Closed Sales (Retention)

- Upon successful conversion, the lead is tagged as "Closed Sales".
 - Data is migrated to the primary CRM database for lifecycle management, focusing on post-sale satisfaction, loyalty rewards, and referral generation, thereby creating a foundation for repeat business and targeted upselling opportunities.
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3. Operational Cost Breakdown (MYR)

Note: This strictly covers recurring infrastructure and software costs. One-time setup and development labor are excluded.

3.1 Operation/Maintenance Cost

Cost Component	Option A: n8n Cloud (Managed)	Option B: Cloud VPS (Self-Hosted)	Option C: Hybrid On-Premise	Option D: Full On-Premise (No Cloud AI)
n8n Hosting	~RM 100 / mo	RM 0 (Community Edition)	RM 0 (Community Edition)	RM 0 (Community Edition)
Cloud AI Tokens	~RM 100 / mo*	~RM 100 / mo*	~RM 100 / mo*	RM 0 (Local Model)
Meta API Fees	RM 0 (Reply < 24h)	RM 0 (Reply < 24h)	RM 0 (Reply < 24h)	RM 0 (Reply < 24h)
Power / Maint.	RM 0	RM 0	~RM 15 / mo	~RM 30 / mo
Executions	2,500 / mo limit	Unlimited	Unlimited	Unlimited
Est. Monthly	~RM 200 / mo	~RM 140 / mo	~RM 115 / mo	~RM 30 / mo

3.2 Hardware Estimated Cost

Hardware Route	Key Component	Estimated Upfront Cost (MYR)	Pros & Cons
Option 1: Apple Silicon	Mac Mini (M-Series) with 32GB Unified Memory	~RM 5,200 - RM 6,500	• Pros: Extremely power-efficient, silent, and easily handles both n8n and the AI model simultaneously. • Cons: Slightly slower token generation compared to dedicated NVIDIA GPUs.
Option 2: Standard AI PC	PC build with an NVIDIA RTX 4070 Ti Super (16GB VRAM)	~RM 7,000 - RM 8,500	• Pros: High-speed inference, natively supports CUDA, excellent for fast WhatsApp routing. • Cons: Higher electricity draw; generates heat and noise.
Option 3: High-Capacity AI PC	PC build with a Used NVIDIA RTX 3090 (24GB VRAM)	~RM 6,500 - RM 8,500	• Pros: The extra 8GB of VRAM gives you headroom to run larger context windows or a second visual AI model alongside n8n. • Cons: Requires a heavy-duty power supply and large PC case; buying used parts carries some risk.

4. Data Architecture & Privacy Schema

4.1 Data Security & Hosting Strategy

Due to the presence of Personally Identifiable Information (PII) and sensitive property documents, relying on third-party cloud spreadsheets (such as Google Sheets or Airtable) introduces unnecessary data governance risks.

This solution utilizes a **Locally Hosted PostgreSQL Database** deployed alongside the n8n orchestration engine.

- **Absolute Privacy:** All client chat histories, intent classifications, and contact details remain strictly on the localized server. n8n queries this data locally, ensuring zero exposure to the open internet.
- **Hybrid Asset Management:** pCloud will serve exclusively as the secure file repository. The local database will only store the encrypted URL links pointing to these public/private assets (such as viewing PDFs and property images).
- **Cost Efficiency:** PostgreSQL is an enterprise-grade, open-source database requiring RM 0 in monthly licensing fees.

5. Project Delivery Timeline

- **Week 1-2:** Meta API provisioning, infrastructure setup, and establishing the sub-number/main-number routing logic.
- **Week 3-4:** Configuring pCloud integrations, multi-channel lead ingestion, and the Phase 1 AI intent classification rules.
- **Week 5-6:** Developing Phase 2 (Google Calendar sync, WhatsApp labeling) and Phase 3 (Acknowledgement PDF generation, automated follow-up scheduling).
- **Week 7-8:** End-to-end testing of the Hot/Warm/Cooling escalation paths, final QA, and handover.

(Subject to changes due to unforeseen circumstances)